Real-time Energy Monitoring and Power Consumption Analysis

Submitted in the partial fulfilment for the Course of

to the award of the degree of

BACHELOR OF ENGINEERING

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COMPUTER SCIENCE AND ENGINEERING

Submitted by

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DECLARATION

We, **M.Nivedha (192471040),**of the Bachelor Of Engineering In Computer Science And Engineering**,** Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled **'Real-time Energy Monitoring and Power Consumption Analysis"**is the result of our own Bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

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ABSTRACT

Real-time Energy Monitoring and Power Consumption Analysis is a proposed system developed to provide continuous and efficient monitoring of electrical energy consumption. Conventional energy meters generally provide limited information and often require manual observation to understand detailed consumption patterns. This project addresses this limitation by integrating electrical parameter measurement, real-time processing, energy calculation, monitoring, and alert generation into a single system.

The proposed system measures important electrical parameters such as voltage and current using suitable sensors. The collected sensor data is transferred to a microcontroller or processing unit for further analysis. Instantaneous electrical power is calculated using the relationship $P = V \times I$, while energy consumption is determined by accumulating power measurements over a period of time. The processed information can be displayed through a suitable monitoring interface, allowing users to observe energy consumption in real time.

The system also supports the analysis of power consumption patterns under different operating conditions. By continuously recording measurement values, variations in power demand, periods of high consumption, and possible unnecessary energy usage can be identified. A threshold-based detection mechanism is incorporated to compare measured consumption with a predefined limit. When the consumption exceeds the specified threshold, the system can generate an alert to provide early notification of excessive power usage.

The project combines energy data acquisition, power and energy calculation, real-time monitoring, consumption analysis, and threshold-based alerts. This integrated approach reduces dependence on manual monitoring and provides clearer information for energy-management decisions. It can help users understand their consumption behaviour and identify opportunities for reducing unnecessary energy usage.

The proposed system is also designed with future expansion in mind. It can be extended using IoT connectivity, cloud-based data storage, mobile applications, remote monitoring, predictive energy analysis, and automated load control. Advanced analysis can further support peak-load identification, consumption forecasting, abnormal usage detection, and energy-cost estimation.

Overall, the project provides a practical engineering approach for real-time electrical energy monitoring and power consumption analysis, with the potential to improve energy awareness, monitoring efficiency, and informed energy-management practices.

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LIST OF ABBREVIATIONS / SYMBOLS

Standard	Description	Application in Project
IEC 62052	Electricity metering equipment requirements	Supports energy meter measurement and safety considerations
IEC 62053	Electricity metering equipment accuracy requirements	Used for considering measurement accuracy
IEC 61010	Safety requirements for measurement equipment	Supports safe design and testing of the monitoring system
IEEE 1459	Definitions for electric power quantities	Supports power and energy measurement calculations
Modbus	Industrial communication protocol	Can be used for communication between meter/sensors and controller
Wi-Fi / IEEE 802.11	Wireless LAN communication standard	Supports wireless monitoring and data transmission
IPv4 / IPv6	Internet Protocol standards	Supports network addressing and communication
WPA2 / WPA3	Wireless security standards	Provides security for wireless energy-monitoring communication

CHAPTER-1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Electrical energy is an essential resource in homes, educational institutions, offices, industries, laboratories, and commercial buildings. The increasing use of electrical appliances and electronic equipment has resulted in higher energy consumption. Conventional energy meters generally provide cumulative energy readings, but they do not always provide detailed real-time information about instantaneous power consumption and usage patterns.

Real-time energy monitoring provides a systematic way to measure, process, and analyse electrical parameters such as voltage, current, power, and energy consumption. By continuously collecting electrical data, users can understand how much energy is being consumed and identify periods of excessive consumption.

The proposed **Real-time Energy Monitoring and Power Consumption Analysis** system uses voltage and current measurements to calculate electrical power and cumulative energy consumption. The measured information can be processed by a microcontroller or processing unit and displayed through a monitoring interface. The system can also compare measured consumption with predefined threshold values and generate alerts when excessive consumption is detected.

The overall approach consists of four major stages: **energy data acquisition, power and energy calculation, energy monitoring and analysis, and threshold-based alert generation**.

1.2 Need for the Project

Traditional energy monitoring methods have several limitations, particularly when users need detailed information about consumption patterns. Manual recording of meter readings can be time-consuming and does not provide continuous monitoring.

The need for the proposed project arises from the following factors:

1. Increasing electrical energy consumption.
2. Lack of continuous real-time monitoring in conventional systems.
3. Difficulty in identifying excessive power consumption at an early stage.
4. Need to analyse energy consumption patterns.
5. Requirement for accurate voltage and current measurements.
6. Need for automatic calculation of power and cumulative energy.
7. Need for immediate alerts when consumption exceeds predefined limits.

8. Support for better energy-management decisions.

Real-time monitoring can therefore help users understand energy usage and identify opportunities for reducing unnecessary consumption.

1.3 Problem Statement

Conventional energy meters mainly provide cumulative energy readings and require users to manually monitor consumption. This makes it difficult to identify sudden increases in power usage, determine consumption patterns, and detect excessive energy consumption immediately.

The engineering problem is therefore to develop a system that can measure electrical parameters in real time, calculate power and energy consumption, monitor usage patterns, and generate alerts when predefined consumption limits are exceeded.

The proposed system addresses this problem by integrating electrical sensors, data acquisition, computational processing, monitoring, analysis, and threshold-based alert generation into a single energy-monitoring framework.

1.4 Project Aim

The aim of this project is to design and develop a real-time energy monitoring system for measuring, analysing, and managing electrical power consumption using continuously acquired voltage and current data.

The system is intended to provide real-time information about electrical consumption and support early detection of excessive energy usage through threshold-based alerts.

1.5 Project Objectives

The main objectives of the project are:

1. To measure voltage and current values using suitable electrical sensors.
2. To acquire electrical measurements continuously.
3. To calculate instantaneous power from measured electrical parameters.
4. To calculate cumulative energy consumption.
5. To monitor power and energy values in real time.
6. To visualize energy consumption data clearly.
7. To analyse energy consumption patterns.
8. To establish predefined power or energy thresholds.
9. To detect excessive energy consumption automatically.
10. To generate alerts when the specified threshold is exceeded.
11. To provide a systematic basis for energy-management decisions.

12. To evaluate the performance and effectiveness of the proposed monitoring system.

1.6 Scope

The scope of the project includes the design and analysis of a real-time electrical energy monitoring system.

The project covers:

- Voltage measurement.
- Current measurement.
- Real-time data acquisition.
- Power calculation.
- Energy calculation.
- Cumulative energy monitoring.
- Consumption visualization.
- Consumption-pattern analysis.
- Threshold configuration.
- Excessive-consumption detection.
- Alert generation.
- Performance evaluation.

The project does not primarily cover:

- Utility-scale electricity-grid management.
- Large-scale industrial power distribution.
- Modification of utility electricity infrastructure.
- Commercial deployment of a complete smart-grid system.
- Long-term field deployment beyond the defined project requirements.

The system can be further extended in future work with cloud monitoring, mobile applications, IoT connectivity, predictive analytics, and intelligent energy-management features.

1.7 Expected Engineering Outcome

The expected outcome is a functional **Real-time Energy Monitoring and Power Consumption Analysis** system capable of acquiring electrical measurements and converting them into meaningful power and energy information.

The expected engineering outcomes are:

\

Expected Outcome	Description
Real-time Electrical Data	Continuous voltage and current measurements
Power Calculation	Determination of instantaneous power consumption
Energy Calculation	Measurement of cumulative energy consumption
Real-time Monitoring	Continuous display of power and energy values
Consumption Analysis	Identification of usage patterns
Threshold Detection	Comparison of consumption with predefined limits
Alert Generation	Notification when excessive consumption occurs
Energy Management	Support for identifying unnecessary or high consumption

Overall, the project is expected to provide a practical framework for **accurate measurement, real-time monitoring, analysis, and early detection of excessive electrical energy consumption**.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review was conducted to understand existing approaches for electrical energy monitoring, power measurement, energy analysis, and consumption-alert systems. The review focused on the methods used to measure electrical parameters, calculate power and energy, monitor consumption in real time, and identify excessive energy usage.

The review considered the following areas:

- Conventional energy-metering methods.
- Digital energy-monitoring systems.
- Sensor-based voltage and current measurement.
- Microcontroller-based energy monitoring.
- IoT-based energy-monitoring systems.
- Real-time power and energy calculation.
- Energy-consumption visualization.
- Threshold-based alert mechanisms.

The reviewed approaches were compared based on **measurement capability, real-time monitoring, analysis, alert generation, scalability, cost, and implementation complexity**.

2.2 Review of Existing Work

2.2.1 Conventional Energy Meter

Traditional energy meters are primarily used to measure cumulative electrical energy consumption. Users generally read the meter periodically to determine energy usage.

Advantages:

- Simple operation.
- Reliable measurement of cumulative consumption.
- Suitable for basic energy billing.

Limitations:

- Limited real-time monitoring.
- Manual observation may be required.
- Difficult to identify sudden increases in consumption.
- Limited analysis of consumption patterns.
- No automatic alert mechanism in basic implementations.

2.2.2 Digital Energy Monitoring Systems

Digital energy meters use electronic measurement techniques to provide electrical measurements in numerical form. They can provide more convenient readings than conventional meters.

Typical parameters include:

- Voltage
- Current
- Power
- Energy consumption
- Power factor

Digital monitoring improves accessibility to electrical information but basic systems may still provide limited analytical and alerting capabilities.

2.2.3 Sensor-Based Energy Monitoring

Sensor-based systems use voltage and current sensors to acquire electrical parameters continuously. The sensor outputs are provided to a processing unit, where the measurements are converted into useful electrical information.

The basic process is:

Voltage/Current Measurement → Data Acquisition → Processing → Power Calculation → Energy Calculation

This approach is suitable for real-time monitoring because measurements can be acquired at regular intervals and processed automatically.

2.2.4 Microcontroller-Based Energy Monitoring

Microcontroller-based systems provide a compact method for processing sensor measurements. The microcontroller can receive voltage and current readings, calculate power and energy, and send the results to a display or monitoring application.

A typical architecture is:

Sensors → Microcontroller → Calculation → Display/Monitoring → Alert

Such systems can reduce manual monitoring and provide continuous information about electrical consumption.

2.2.5 IoT-Based Energy Monitoring

IoT-based energy-monitoring systems extend conventional monitoring by transmitting energy data through a communication network. The collected information can be viewed remotely using a web-based dashboard or mobile application.

The general architecture is:

Energy Sensors → Processing Unit → Communication Network → Cloud/Server → User Dashboard

IoT integration can provide remote monitoring and historical data analysis. However, it may introduce additional requirements such as network connectivity, data storage, security, and communication infrastructure.

2.3 Comparative Analysis

The existing approaches can be compared with the proposed real-time energy monitoring system.

Parameter	Conventional Meter	Digital Meter	Sensor-Based System	Proposed System
Voltage Measurement	Limited	Yes	Yes	**Yes**
Current Measurement	Limited	Yes	Yes	**Yes**
Real-Time Monitoring	Low	Moderate	High	**High**
Power Calculation	Limited	Yes	Yes	**Yes**
Energy Calculation	Yes	Yes	Yes	**Yes**
Consumption Analysis	Low	Moderate	Moderate	**High**
Visualization	Limited	Basic	Possible	**Yes**
Threshold Detection	No/limited	Limited	Possible	**Yes**
Automatic Alerts	No	Limited	Possible	**Yes**
Manual Monitoring	Required	Reduced	Reduced	**Minimized**

The comparison indicates that the proposed system combines **real-time measurement, power and energy calculation, monitoring, consumption analysis, and threshold-based alerts** within one framework.

2.4 Research / Engineering Gap

The review of existing approaches identifies several engineering gaps.

1. Limited Real-Time Information

Traditional systems may primarily provide cumulative energy readings rather than continuously presenting detailed consumption information.

2. Lack of Consumption Analysis

Basic energy meters do not necessarily provide tools for analysing consumption patterns or identifying periods of high energy usage.

3. Delayed Detection of Excessive Consumption

Without automatic threshold monitoring, users may become aware of unusually high consumption only after reviewing accumulated readings.

4. Limited Integration

Some basic monitoring solutions focus only on measurement and do not combine measurement, calculation, visualization, analysis, and alert generation.

5. Manual Monitoring

Manual meter reading requires user involvement and makes continuous observation difficult. Therefore, there is a need for an integrated system capable of **acquiring electrical data continuously, processing the measurements, analysing consumption, and providing alerts when predefined limits are exceeded**.

2.5 Proposed Contribution

The proposed **Real-time Energy Monitoring and Power Consumption Analysis** system addresses the identified engineering gaps by integrating four major functional modules:

Module 1 – Energy Data Acquisition

Voltage and current sensors are used to collect electrical measurements continuously.

Module 2 – Power & Energy Calculation

The acquired measurements are processed to calculate instantaneous power and cumulative energy consumption.

Module 3 – Energy Monitoring & Analysis

The calculated values are displayed and analysed to identify energy-consumption patterns and variations.

Module 4 – Threshold & Alert System

Predefined consumption limits are established. The measured values are compared with these limits, and an alert is generated when excessive consumption is detected.

Proposed System Flow

Energy Sensors

↓

Data Acquisition

↓

Microcontroller / Processing Unit

↓

Power & Energy Calculation

↓

Real-Time Monitoring

↓

Consumption Analysis

↓

Threshold Detection

↓

Alert Generation

The proposed contribution is therefore an integrated energy-monitoring approach that moves beyond simple meter reading by providing **real-time electrical data, automated calculations, consumption analysis, and early detection of excessive power consumption**.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

The proposed **Real-time Energy Monitoring and Power Consumption Analysis** system is designed to continuously measure electrical parameters, calculate power and energy consumption, analyse consumption patterns, and provide alerts when consumption exceeds predefined limits. The requirements are divided into **functional, performance, and system requirements**.

3.1.1 Functional Requirements

The system should:

1. Measure electrical voltage continuously.
2. Measure electrical current continuously.
3. Acquire sensor readings in real time.
4. Process the measured electrical data.
5. Calculate instantaneous power consumption.
6. Calculate cumulative energy consumption.
7. Display real-time electrical values.
8. Store or record energy-consumption data for analysis.
9. Analyse power-consumption patterns.
10. Compare measured consumption with predefined thresholds.
11. Detect excessive power or energy consumption.
12. Generate an alert when the specified threshold is exceeded.
13. Provide a basis for identifying unnecessary energy consumption.
14. Support energy-management decisions.

3.1.2 Performance Requirements

The system is evaluated using the following parameters:

Parameter	Requirement
Voltage	Accurate measurement of electrical voltage
Current	Accurate measurement of load current
Power	Should provide reliable instantaneous power values
Energy	Should calculate cumulative energy consumption
Sampling	Should acquire readings at suitable intervals
Response Time	Should provide monitoring with minimum delay
Accuracy	Measurements should be sufficiently accurate for the selected application
Monitoring	Should provide continuous consumption information
Threshold Detection	Should identify excessive consumption promptly
Alert	Should generate an alert when the defined limit is exceeded

The actual numerical accuracy and response-time targets should be defined according to the selected sensors, hardware, and experimental setup rather than assuming values without measurement.

3.2 Constraints & Applicable Standards

3.2.1 Project Constraints

The following constraints are considered during system design:

- Sensor measurement accuracy.
- Sensor operating range.
- Electrical safety requirements.
- Limited availability of physical energy-consumption data.
- Sampling and processing limitations.
- Microcontroller processing capability.
- Data-storage limitations.
- Communication limitations, where wireless monitoring is used.
- Cost of sensors and hardware components.
- Variations in electrical loads.
- Calibration requirements.
- Environmental effects on measurement accuracy.
- Availability of suitable monitoring equipment.

3.2.2 Applicable Standards

The proposed system should follow applicable electrical measurement, safety, and communication practices.

Important considerations include:

- Electrical safety requirements.
- Appropriate sensor ratings.
- Safe isolation between measurement circuits and low-voltage processing circuits.
- Proper grounding and wiring practices.
- Manufacturer specifications for sensors and electronic components.
- Applicable institutional electrical-safety procedures.
- Secure communication practices when monitoring data is transmitted through a network.

The selected components should be operated within their specified voltage, current, temperature, and measurement limits.

3.3 Design Specifications

The proposed energy-monitoring system consists of electrical sensors, a data-acquisition unit, processing hardware, a monitoring interface, and a threshold-based alert mechanism.

3.3.1 Basic Architecture

The proposed architecture can be represented as:

Electrical Load → Voltage & Current Sensors → Data Acquisition → Microcontroller / Processing Unit → Power & Energy Calculation → Monitoring Interface → Alert System

The sensors measure the electrical parameters associated with the load. The processing unit receives the measurements and calculates power and energy consumption.

3.3.2 Design Parameters

Design Parameter	Description
Voltage	Represents the electrical potential measured across the load
Current	Represents the current drawn by the load
Power	Represents instantaneous electrical power consumption
Energy	Represents cumulative electrical energy consumption
Sampling Interval	Defines how frequently measurements are collected
Threshold Value	Defines the maximum permitted consumption
Alert Status	Indicates whether consumption exceeds the threshold

Design Parameter	Description
Time	Records the time associated with each measurement
Consumption Pattern	Represents variation in energy usage over time

3.3.3 Energy Monitoring Design Considerations

The design considers:

- Accurate voltage measurement.
- Accurate current measurement.
- Appropriate sensor selection.
- Regular data acquisition.
- Reliable power calculation.
- Cumulative energy calculation.
- Real-time display.
- Historical consumption analysis.
- Threshold configuration.
- Alert generation.
- Safe electrical connections.
- Reliable data storage.

The design should ensure that the monitoring system does not interfere with normal operation of the electrical load.

3.4 Alternative Solutions & Evaluation

Different approaches can be considered before selecting the proposed real-time monitoring system.

Alternative	Advantages	Limitations	Suitability
Conventional Energy Meter	Simple and widely available	Limited real-time analysis	Low
Digital Energy Meter	Provides digital readings	Limited analysis in basic models	Moderate
Manual Energy Monitoring	Low implementation cost	Time-consuming and delayed	Low
Sensor-Based Monitoring	Real-time electrical measurements	Requires sensors and processing	High

Alternative	Advantages	Limitations	Suitability
IoT-Based Monitoring	Remote monitoring and data access	Requires communication infrastructure	High
Proposed System	Measurement, calculation, analysis and alerts	Requires integrated hardware/software	**High**

Evaluation

Conventional meters are useful for measuring accumulated energy but provide limited information about real-time consumption patterns.

Digital meters improve the readability of electrical measurements but may not provide advanced analysis or automatic alerts.

Manual monitoring requires repeated observation and does not support continuous monitoring.

Sensor-based monitoring provides the ability to acquire voltage and current measurements continuously. When combined with processing and visualization, it can provide detailed information about power and energy consumption.

IoT-based monitoring can provide remote access to energy data but introduces additional communication and security requirements.

Therefore, the proposed integrated system is selected because it combines **real-time measurement, calculation, monitoring, analysis, and threshold-based alert generation**.

3.5 Selected Design & Detailed Design

3.5.1 Selected Design

The selected design integrates four major modules:

Energy Data Acquisition

↓

Power & Energy Calculation

↓

Energy Monitoring & Analysis

↓

Threshold & Alert System

The complete conceptual design is:

Electrical Load

↓

Voltage & Current Sensors

↓

Data Acquisition

↓

Microcontroller / Processing Unit

↓

Power & Energy Calculation

↓

Real-Time Monitoring

↓

Consumption Analysis

↓

Threshold Detection

↓

Alert Generation

3.5.2 Energy Data Acquisition

Voltage and current sensors are used to collect electrical measurements.

The data-acquisition module performs the following operations:

1. Measure voltage.
2. Measure current.
3. Convert sensor outputs into usable measurement values.
4. Collect readings at regular intervals.
5. Transfer the measurements to the processing unit.

Accurate data acquisition is important because errors in the measured voltage or current can affect the calculated power and energy values.

3.5.3 Power & Energy Calculation

The measured electrical parameters are processed to determine power consumption.

For a basic electrical load:

Power:

$$P = V \times I$$

where:

- **P** = Power in watts (W)
- **V** = Voltage in volts (V)

- **I** = Current in amperes (A)

Energy consumption can be calculated by accumulating power over time:

$$E = \int P\,(t)\,dt$$

For discrete measurements, the system can approximate energy using the measured power and sampling interval.

Energy is commonly represented in **watt-hours (Wh)** or **kilowatt-hours (kWh)**.

3.5.4 Real-Time Monitoring

The calculated values are presented through a monitoring interface.

The interface can display:

- Voltage.
- Current.
- Instantaneous power.
- Cumulative energy.
- Time.
- Consumption status.
- Alert status.

This allows users to observe changes in electrical consumption without relying entirely on manual meter readings.

3.5.5 Energy Consumption Analysis

The collected data can be analysed to identify:

- High-consumption periods.
- Low-consumption periods.
- Sudden increases in power usage.
- Daily consumption patterns.
- Changes in load behaviour.
- Excessive energy consumption.

Graphs and tables can be used to improve understanding of consumption patterns.

3.5.6 Threshold & Alert System

A predefined power or energy threshold is established.

The basic operation is:

Measured Value → Compare with Threshold → Threshold Exceeded?

If **No**:

Continue Monitoring

If **Yes**:

Generate Alert

The alert mechanism can indicate that the current power consumption or accumulated energy consumption has exceeded the specified limit.

3.6 Trade-off Analysis

The energy-monitoring system involves several engineering trade-offs. Improving one design parameter may affect another.

3.6.1 Accuracy vs Cost

Higher-accuracy sensors can improve measurement quality but may increase system cost.

Design decision: Select sensors that provide sufficient accuracy for the project while maintaining reasonable implementation cost.

3.6.2 Sampling Rate vs Processing Requirement

Increasing the sampling frequency provides more measurement information but increases processing and data-storage requirements.

Design decision: Select a sampling interval that provides sufficient monitoring detail without unnecessary processing overhead.

3.6.3 Real-Time Monitoring vs Data Storage

Continuous monitoring generates large amounts of measurement data.

Design decision: Store the measurements required for meaningful analysis while avoiding unnecessary data accumulation

3.6.4 Sensor Range vs Measurement Resolution

A sensor with a large measurement range can support larger loads, but measurement resolution may be affected depending on the sensor design.

Design decision: Select a sensor range appropriate for the expected electrical load.

3.6.5 Monitoring Features vs System Complexity

Adding dashboards, data storage, communication, and alerts improves functionality but increases system complexity.

Design decision: Include features that directly support real-time monitoring and energy-consumption analysis.

Trade-off Summary

Design Factor	**Increasing It**	**Possible Benefit**	**Possible Disadvantage**
Sensor Accuracy	Higher	More accurate measurements	Higher cost
Sampling Rate	Higher	More detailed monitoring	Higher processing requirement
Data Storage	Higher	More historical information	More storage requirement
Sensor Range	Higher	Supports larger loads	May affect resolution
Monitoring Features	More	Better analysis	Greater complexity
Communication	More frequent	Faster remote updates	Higher network/resource usage

3.7 Sustainability, Ethics, Safety, Risk & Security

3.7.1 Sustainability

Sustainability is an important consideration because the primary objective of the project is to understand and manage electrical energy consumption.

The proposed system supports sustainability through:

- Identification of unnecessary energy consumption.
- Real-time monitoring of electrical usage.
- Detection of excessive power consumption.
- Improved awareness of energy usage.
- Support for energy-saving decisions.
- Reduced dependence on manual monitoring.
- Efficient use of electrical resources.

A monitoring system can help users identify consumption patterns and take appropriate actions to reduce unnecessary energy usage.

3.7.2 Ethical Considerations

The project should follow responsible engineering practices.

Important ethical considerations include:

- Reporting measurement results honestly.
- Avoiding fabrication or manipulation of experimental data.
- Maintaining accuracy in calculations.
- Protecting collected energy-consumption data.
- Collecting only necessary information.
- Avoiding unauthorized access to monitoring information.
- Clearly documenting assumptions and limitations.
- Using experimental results responsibly.

The data generated by the system should be used only for legitimate monitoring and analysis purposes.

3.7.3 Safety Considerations

Electrical energy monitoring requires particular attention to safety because the system may interact with electrical circuits.

Safety considerations include:

- Using appropriately rated voltage and current sensors.
- Maintaining proper electrical isolation.
- Avoiding direct contact with energized conductors.
- Using insulated wiring and suitable connectors.
- Following manufacturer instructions.
- Providing proper enclosure for electronic components.
- Preventing short circuits.
- Avoiding overloaded components.
- Ensuring safe grounding where applicable.
- Disconnecting power before modifying electrical connections.

Only appropriate and properly rated components should be used for the intended electrical load.

3.7.4 Risk Analysis

Risk	Possible Effect	Mitigation
Sensor failure	Incorrect measurements	Use suitable and rated sensors
Incorrect calibration	Measurement error	Calibrate sensors before testing
Excessive voltage/current	Component damage	Select appropriate sensor ratings
Loose connection	Unstable readings	Check and secure connections
Electrical fault	Safety hazard	Use proper isolation and protection
Incorrect calculation	Wrong power/energy values	Verify calculation algorithms
Data loss	Loss of consumption history	Use suitable data-storage methods
Threshold misconfiguration	Incorrect alerts	Verify threshold settings
Communication failure	Delayed monitoring	Monitor local values and verify communication
Software error	Incorrect analysis	Test and validate the software
Sensor noise	Fluctuating readings	Apply suitable filtering/calibration

3.7.5 System Security

If the monitoring system stores or transmits energy data, appropriate security measures should be considered.

Security measures include:

- Authentication for monitoring interfaces.
- Access control.
- Secure communication where applicable.
- Protection of stored energy data.
- Restriction of administrative access.
- Regular software updates.
- Protection against unauthorized modification.
- Backup of important monitoring data.

Only authorized users should be permitted to configure thresholds, modify system settings, or access sensitive monitoring information.

Chapter 3 Summary

This chapter presented the engineering design of the proposed Real-time Energy Monitoring and Power Consumption Analysis system. The functional and performance requirements, project constraints, design specifications, alternative solutions, and selected architecture were identified.

The proposed system integrates energy data acquisition, power and energy calculation, real-time monitoring, consumption analysis, and threshold-based alert generation. Important trade-offs involving sensor accuracy, cost, sampling rate, processing requirements, data storage, and system complexity were also considered.

Finally, sustainability, ethical practices, electrical safety, risk management, and system security were incorporated into the overall design. These design decisions provide the foundation for the implementation, testing, and results presented in Chapter 4.

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The proposed **Real-time Energy Monitoring and Power Consumption Analysis** system was developed using a systematic approach consisting of energy data acquisition, power and energy calculation, real-time monitoring, consumption analysis, and threshold-based alert generation. The system first acquires voltage and current measurements from the electrical load using appropriate sensors. These measurements are processed to calculate instantaneous power and cumulative energy consumption. The processed information is then presented through a monitoring interface to support analysis of consumption patterns and detection of excessive energy usage.

Resources Used

Resource	Purpose
Voltage Sensor	Measurement of electrical voltage
Current Sensor	Measurement of load current
Microcontroller / Processing Unit	Data acquisition and processing
Monitoring Interface	Display of electrical parameters
Computer	Programming, testing and analysis
Software / Simulation Tool	System implementation and verification
Data Storage	Recording energy consumption values
Connecting Wires	Electrical and sensor connections
Power Supply	Providing required operating power

The development process follows the sequence:

Sensor Selection → Data Acquisition → Data Processing → Power Calculation → Energy Calculation → Monitoring → Consumption Analysis → Threshold Detection → Alert Generation

4.2 Software / Simulation Implementation & Modern Tools Used

The implementation consists of data acquisition, calculation, monitoring, and analysis components. The software processes the voltage and current values obtained from the measurement system and calculates electrical parameters in real time.

4.2.1 Data Acquisition Implementation

The voltage and current sensors are connected to the processing unit. The processing unit periodically reads the sensor outputs and converts them into usable electrical measurements.

The acquired parameters include:

- Voltage (V)
- Current (A)
- Instantaneous Power (W)
- Cumulative Energy (Wh/kWh)
- Time
- Threshold status
- Alert status

The acquired readings are processed continuously so that changes in the electrical load can be observed.

4.2.2 Power Calculation

The instantaneous electrical power is calculated using:

$$P = V \times I$$

where:

- P= Power in watts (W)
- V= Voltage in volts (V)
- I= Current in amperes (A)

For example, when the measured voltage and current change, the calculated power also changes accordingly.

4.2.3 Energy Calculation

Energy consumption is calculated from the measured power over time.

$$E = \int P\,(t)\,dt$$

For sampled measurements, the energy can be estimated using:

$$E \approx \sum P_i\,\Delta t$$

where $P_i$represents the measured power and Δtrepresents the sampling interval.

The resulting energy value can be represented in **Wh or kWh**.

4.2.4 Real-Time Monitoring

The monitoring interface displays important parameters such as:

- Voltage
- Current

- Power
- Cumulative energy
- Time
- Consumption status
- Threshold status
- Alert condition

This allows users to observe energy consumption without depending entirely on manual meter readings.

4.2.5 Tools Used

Tool / Software / Equipment	Purpose	Stage Used
Microcontroller / Processing Unit	Sensor data processing	Implementation
Voltage Sensor	Voltage measurement	Data acquisition
Current Sensor	Current measurement	Data acquisition
Programming Environment	Program development	Implementation
Simulation Environment	Functional verification	Testing
Spreadsheet / Data Analysis Tool	Data organization and analysis	Results
Monitoring Interface	Real-time display	Monitoring

The reference structure recommends listing only tools that have a purposeful role in the project rather than providing a generic software list.

Figure 4.1: Energy Monitoring System Implementation

Figure 4.2: Data Acquisition and Processing Program

Figure 4.3: Real-Time Energy Monitoring Interface

4.3 Test Plan & Verification

Testing was planned to verify whether the proposed system performs the required functions correctly. The test plan covers sensor measurement, power calculation, energy calculation, real-time monitoring, consumption analysis, and threshold alert generation.

The reference report specifies that the test plan should be defined before presenting the results and should include the requirement, test method, acceptance criterion, achieved value and Pass/Fail status.

4.3.1 Functional Test Plan

Requirement	Test Method	Acceptance Criterion	Achieved Value	Status
Voltage measurement	Apply electrical load and observe sensor reading	Voltage value should be detected	To be measured	Pass/Fail
Current measurement	Connect load and observe current	Current value should be detected	To be measured	Pass/Fail
Power calculation	Compare calculated power with $V \times I$	Correct power calculation	To be measured	Pass/Fail
Energy calculation	Monitor load for a known period	Energy should accumulate correctly	To be measured	Pass/Fail
Real-time display	Change load and observe interface	Values should update	To be measured	Pass/Fail
Data recording	Record successive measurements	Data should be stored correctly	To be measured	Pass/Fail
Consumption analysis	Analyse recorded measurements	Consumption pattern should be identifiable	To be measured	Pass/Fail
Threshold detection	Apply value above threshold	Excessive value should be detected	To be measured	Pass/Fail
Alert generation	Trigger threshold condition	Alert should be generated	To be measured	Pass/Fail

4.3.2 Verification Procedure

1. Switch on the monitoring system.
2. Connect the required electrical load.
3. Verify voltage sensor readings.
4. Verify current sensor readings.
5. Calculate instantaneous power.
6. Monitor cumulative energy.
7. Observe the real-time monitoring interface.
8. Record the measured values.
9. Change the electrical load.
10. Observe the change in power consumption.

11. Compare consumption with the predefined threshold.
12. Verify alert generation when the threshold is exceeded.
13. Record the test observations.
14. Analyse the collected data.

4.4 Results

The results of the proposed system are evaluated based on its ability to acquire electrical data, calculate power and energy, display measurements, analyse consumption patterns, and identify excessive energy usage.

4.4.1 Energy Measurement Results

Test Condition	Voltage (V)	Current (A)	Power (W)	Energy (Wh)	Status
Load Condition 1	[Value]	[Value]	[Value]	[Value]	Normal
Load Condition 2	[Value]	[Value]	[Value]	[Value]	Normal
Load Condition 3	[Value]	[Value]	[Value]	[Value]	High
Load Condition 4	[Value]	[Value]	[Value]	[Value]	High

Note: Actual measured values should be entered from the experimental or simulation output rather than using assumed values.

4.4.2 Monitoring Results

The monitoring system provides continuous information about the electrical load. When the load changes, corresponding changes in current and power can be observed. The cumulative energy value increases with operating time and power consumption.

The monitoring output can therefore be used to:

- Identify high-power operating conditions.
- Observe changes in load demand.
- Track cumulative energy consumption.
- Identify periods of excessive usage.
- Support energy-saving decisions.

4.4.3 Threshold and Alert Results

The threshold mechanism compares the measured power or energy value with a predefined limit.

Measured Energy Value → Threshold Comparison → Normal / Excessive → Alert Generation

If the measured value remains within the permitted limit, the system continues normal monitoring.

If the measured value exceeds the threshold, the system generates an alert to indicate excessive consumption.

Suggested Figures

Figure 4.4: Voltage and Current Measurement

Figure 4.5: Power Consumption vs Time

Figure 4.6: Cumulative Energy Consumption

Figure 4.7: Threshold Detection and Alert Output

The source report similarly places measurement graphs and key data under the Results section.

4.5 Data Analysis & Engineering Judgment

The collected voltage, current, power, and energy measurements are analysed to determine the behaviour of the monitored electrical load.

The relationship between voltage, current, and power is particularly important because:

$$P = V \times I$$

Therefore, an increase in current under approximately constant voltage generally results in an increase in power consumption. Continuous monitoring makes these changes easier to identify than manual periodic readings.

Engineering Observations

1. **Voltage Variation:** Changes in voltage can affect the calculated power value.
2. **Current Variation:** Increased load demand generally results in increased current.
3. **Power Variation:** Power changes according to the measured voltage and current.
4. **Energy Accumulation:** Energy consumption increases with operating time and power demand.
5. **Consumption Patterns:** Recorded measurements can reveal high- and low-consumption periods.
6. **Threshold Detection:** Threshold comparison helps identify potentially excessive consumption.
7. **Data-Based Decisions:** Historical measurements can support better energy-management decisions.

Engineering Judgment

The proposed system provides greater visibility into energy usage because it combines measurement, calculation, monitoring, analysis, and alert generation. However, the quality of the final results depends on sensor accuracy, calibration, sampling interval, electrical connection quality, and the reliability of the processing system.

Where measured values differ from expected values, possible causes include sensor tolerance, calibration error, electrical noise, load variation, and measurement limitations. These factors should be considered before making engineering conclusions.

The reference structure specifically requires interpretation of statistical, error, accuracy, or performance measures where applicable and explanation of deviations from expectations.

4.6 Comparison with Existing Solutions & Achievement of Objectives

The proposed system is compared with conventional energy monitoring approaches based on real-time measurement, analysis, and alerting capability.

Objective	Evidence	Achievement
Measure voltage	Voltage sensor measurements	Fully / Partially
Measure current	Current sensor measurements	Fully / Partially
Calculate power	$P = V \times I$ implementation	Fully / Partially
Calculate energy	Cumulative energy calculation	Fully / Partially
Provide real-time monitoring	Monitoring interface	Fully / Partially
Analyse consumption	Recorded energy data	Fully / Partially
Detect excessive consumption	Threshold comparison	Fully / Partially
Generate alerts	Alert mechanism	Fully / Partially
Support energy management	Consumption information	Fully / Partially

Comparison with Existing Methods

Feature	Conventional Meter	Manual Monitoring	Proposed System
Voltage Measurement	Yes	Yes	Yes
Current Measurement	Limited/available	Manual	Yes
Real-Time Monitoring	Limited	No	Yes
Power Calculation	Basic	Manual	Yes
Energy Analysis	Limited	Time-consuming	Yes
Historical Data	Limited	Manual records	Yes
Threshold Detection	Limited	No	Yes
Automatic Alert	Usually limited	No	Yes
Consumption Pattern Analysis	Limited	Difficult	Yes

The proposed solution therefore provides an integrated approach rather than only displaying a meter reading.

4.7 Strengths & Limitations

The reference report requires limitations to be stated honestly rather than presenting the system as perfect.

Strengths

- Provides real-time electrical measurements.
- Measures voltage and current continuously.
- Calculates instantaneous power.
- Calculates cumulative energy consumption.
- Provides a clear monitoring interface.
- Supports consumption-pattern analysis.
- Detects excessive consumption through threshold comparison.
- Can generate alerts for abnormal or high consumption.
- Reduces dependence on manual energy monitoring.
- Supports energy-management decisions.
- Can be extended with IoT or cloud-based monitoring.

Limitations

- Measurement accuracy depends on sensor quality and calibration.
- Electrical installation requires appropriate safety precautions.
- Sensor range may limit the loads that can be monitored.
- High-frequency sampling can increase processing and storage requirements.
- Sensor noise may affect readings.
- Communication failure can affect remote monitoring if wireless connectivity is used.
- The system requires appropriate power and hardware components.
- Long-term operation may require periodic calibration and maintenance.
- Actual energy-saving benefits depend on how users respond to the monitoring information.

4.8 Project Planning, Roles & Budget

The project was divided into stages so that system development, testing, analysis, and documentation could be completed systematically. The reference report recommends including a Gantt chart or milestone table, contribution table, and cost table in this section.

4.8.1 Project Milestone Plan

Phase	Activity	Duration
Phase 1	Topic selection and problem identification	Week 1
Phase 2	Literature review	Week 2
Phase 3	Requirement identification	Week 3
Phase 4	System design	Week 4
Phase 5	Sensor and processing implementation	Week 5
Phase 6	Software / simulation implementation	Week 6
Phase 7	Testing and verification	Week 7
Phase 8	Results and data analysis	Week 8
Phase 9	Report preparation	Week 9
Phase 10	Final review and presentation	Week 10

4.8.2 Responsibility Table

Student	Assigned Responsibility	Actual Contribution
Student 1	System design	Requirements and architecture
Student 2	Data acquisition	Sensor and measurement implementation
Student 3	Software implementation	Processing and monitoring
Student 4	Testing and analysis	Test results and data analysis

If this is an individual project, replace the table with the student's actual responsibilities.

4.8.3 Estimated Budget

Item	Estimated Cost	Actual Cost
Microcontroller	₹[Value]	₹[Value]
Voltage Sensor	₹[Value]	₹[Value]
Current Sensor	₹[Value]	₹[Value]
Display / Monitoring Unit	₹[Value]	₹[Value]
Power Supply	₹[Value]	₹[Value]
Connecting Wires / Components	₹[Value]	₹[Value]
Software / Simulation	₹[Value]	₹[Value]

Item	Estimated Cost	Actual Cost
Other Components	₹[Value]	₹[Value]
Total	**₹[Value]**	**₹[Value]**

Chapter 4 Summary

This chapter presented the implementation, testing, results, analysis, and project management of the **Real-time Energy Monitoring and Power Consumption Analysis** system. The implementation focuses on acquiring voltage and current data, calculating power and cumulative energy, displaying real-time information, analysing consumption patterns, and generating threshold-based alerts. The testing plan provides a structured method for verifying each requirement, while the results and engineering analysis are used to evaluate system performance. The comparison with existing approaches demonstrates the benefit of integrating measurement, calculation, monitoring, analysis, and alert generation into a single system. The strengths and limitations identify both the practical benefits and technical constraints of the proposed solution. Finally, the project planning, responsibilities, and budget provide a structured overview of project execution.

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The project **“Real-time Energy Monitoring and Power Consumption Analysis”** was developed to address the limitations of conventional and manual energy monitoring methods. The proposed system provides a structured approach for measuring electrical parameters, calculating power and energy consumption, monitoring usage in real time, analysing consumption patterns, and identifying excessive energy usage.

The system acquires voltage and current measurements through suitable sensors and processes the collected data using a processing unit. Instantaneous power is calculated using the measured voltage and current, while cumulative energy consumption is determined over time. The monitoring interface provides users with clear information about electrical consumption and helps identify changes in load behaviour.

The threshold and alert mechanism further improves the usefulness of the system by identifying consumption values that exceed predefined limits. This enables early detection of excessive energy usage and supports appropriate energy-management decisions.

Overall, the proposed system provides an integrated solution combining **energy data acquisition, power and energy calculation, real-time monitoring, consumption analysis, and threshold-based alerts**. The project demonstrates how real-time electrical data can be transformed into useful information for improving energy awareness and supporting more efficient energy usage.

5.2 Future Work

The proposed system can be further enhanced to improve its accuracy, scalability, automation, and usability.

5.2.1 IoT-Based Remote Monitoring

The system can be connected to an IoT platform so that energy consumption can be monitored remotely through a web or mobile application.

5.2.2 Cloud-Based Data Storage

Energy measurements can be stored in cloud databases to maintain long-term historical records and provide access from different locations.

5.2.3 Mobile Application

A dedicated mobile application can be developed to display:

- Real-time power consumption
- Daily energy usage
- Weekly and monthly consumption
- Energy-cost estimates
- Threshold alerts
- Historical consumption graphs

5.2.4 AI-Based Consumption Prediction

Machine-learning techniques can be introduced to analyse historical energy data and predict future consumption. This could help identify unusual consumption patterns before they become significant problems.

5.2.5 Automatic Energy Control

The system could be extended to automatically control selected electrical loads when consumption exceeds a predefined limit. This could help reduce unnecessary energy usage.

5.2.6 Improved Sensor Accuracy

Higher-accuracy sensors and improved calibration methods can be used to reduce measurement errors and improve the reliability of energy calculations.

5.2.7 Advanced Analytics

Future versions could provide additional analytical features such as:

- Peak-load identification
- Daily and monthly comparisons
- Energy consumption forecasting
- Abnormal-load detection
- Appliance-level analysis
- Energy-cost analysis
- Efficiency recommendations

5.2.8 Renewable Energy Integration

The system could also be extended to monitor renewable sources such as solar photovoltaic systems and compare generated energy with consumed energy.

5.3 Professional Learning & Individual Reflection

5.3.1 New Knowledge Acquired

Through this project, the following technical knowledge was developed:

- Understanding of electrical energy monitoring systems.
- Knowledge of voltage and current measurement.
- Understanding of power and energy calculations.
- Knowledge of sensor-based data acquisition.
- Understanding of real-time data processing.
- Knowledge of threshold-based monitoring.
- Understanding of energy-consumption analysis.
- Experience with system testing and verification.

5.3.2 Engineering & Professional Skills Developed

The project helped develop:

- Engineering problem-solving skills.
- System design and architecture skills.
- Data analysis skills.
- Programming and implementation skills.
- Testing and debugging skills.
- Technical documentation skills.
- Project planning and time-management skills.
- Engineering decision-making skills.

5.3.3 Individual Reflection

Working on the **Real-time Energy Monitoring and Power Consumption Analysis** project provided practical experience in converting an engineering problem into a structured technical solution. One of the important challenges was ensuring that measured electrical data could be processed correctly to obtain meaningful power and energy values.

A key engineering decision was to integrate **real-time measurement, calculation, monitoring, analysis, and threshold detection** rather than relying only on displaying energy readings. This approach provides more useful information for identifying excessive consumption.

The project also improved my understanding of sensor-based systems, data processing, testing, and engineering documentation. If the project were redesigned, I would focus further on sensor calibration, automated data storage, IoT connectivity, and predictive energy analysis.

Chapter 5 Summary

This chapter concluded the development of the **Real-time Energy Monitoring and Power Consumption Analysis** system. The project provides a practical method for monitoring electrical consumption and identifying excessive usage. Future enhancements such as IoT connectivity, cloud storage, mobile monitoring, AI-based prediction, and automatic load control can further improve the system. The project also provided valuable technical and professional learning in system design, implementation, testing, data analysis, and engineering decision-making.

APPENDICES

Appendix A – Detailed Calculations

This appendix contains the calculations used for power and energy analysis.

A.1 Power Calculation

$$P = V \times I$$

Where:

- P= Electrical power in watts (W)
- V= Voltage in volts (V)
- I= Current in amperes (A)

A.2 Energy Calculation

$$E = P \times t$$

For multiple measurements:

$$E \approx \sum P_i \Delta t$$

Energy may be expressed in **Wh or kWh**.

A.3 Sample Calculation

Parameter	Value
Voltage	[Actual value] V
Current	[Actual value] A
Power	[V × I] W
Operating Time	[Actual value] h
Energy	[P × t] Wh

Note: Replace the bracketed values with actual experimental or simulation measurements.

Appendix B – Engineering Drawings / Schematics

B.1 Overall System Block Diagram

Electrical Load → Voltage & Current Sensors → Data Acquisition → Microcontroller → Power & Energy Calculation → Monitoring Interface → Consumption Analysis → Threshold Detection → Alert

B.2 Sensor Connection Diagram

Show the connections between:

- Voltage sensor
- Current sensor
- Microcontroller
- Power supply
- Electrical load

B.3 Monitoring System Architecture

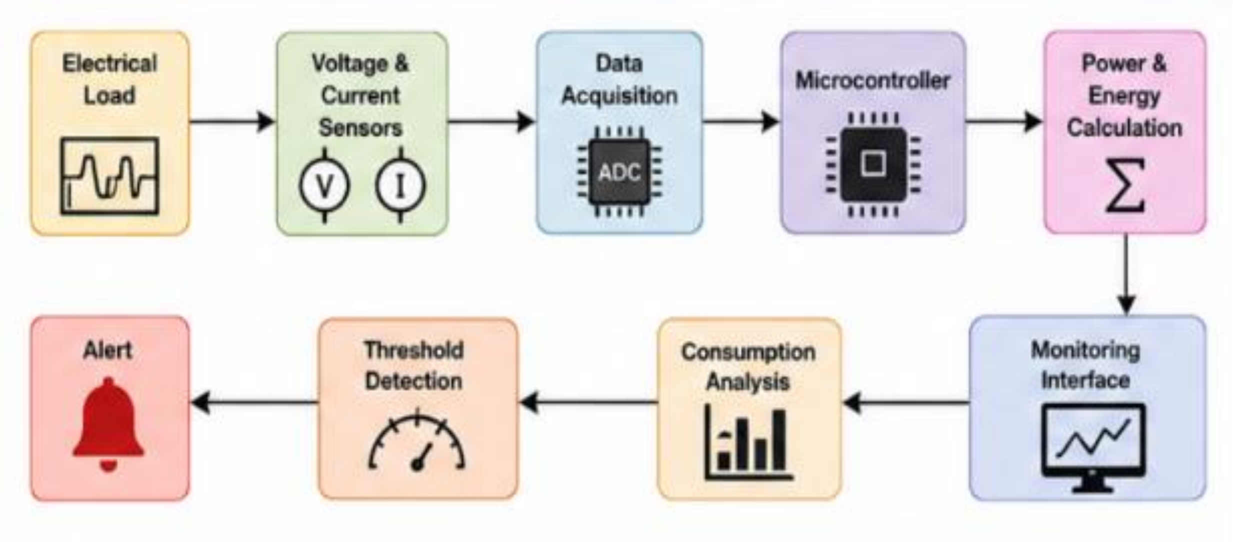

Figure B.1: Proposed Energy Monitoring System Block Diagram

Show the flow of measured data from the sensors to the processing unit and monitoring interface

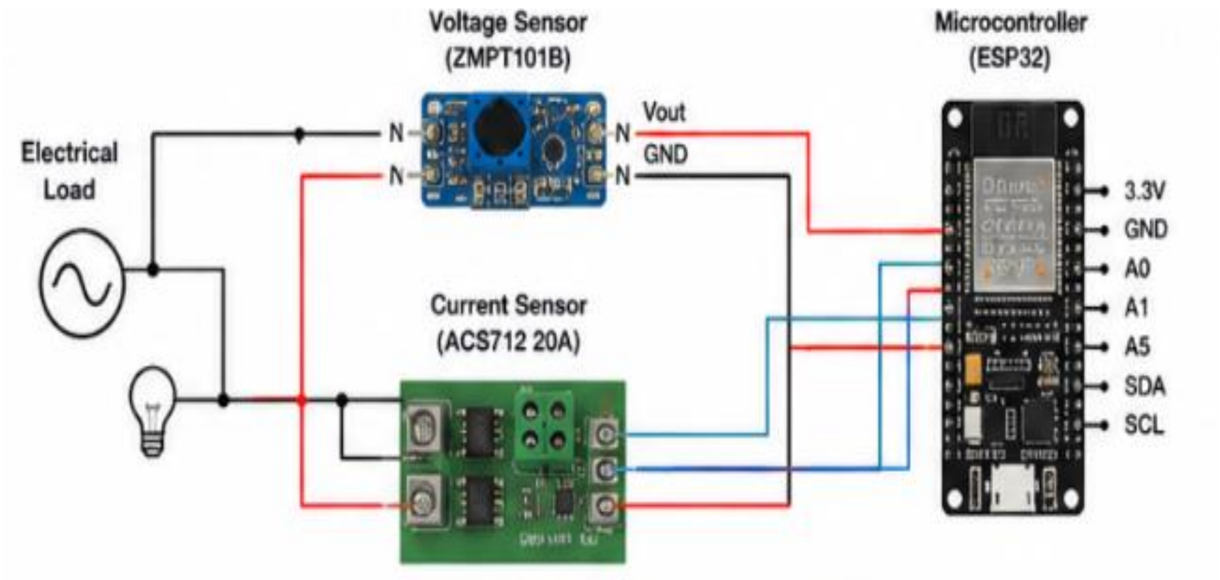

Figure B.2: Sensor and Microcontroller Connection Diagram

Appendix C – Source Code / Code Repository Information

This appendix contains the important source-code sections used for:

- Sensor data acquisition
- Voltage measurement
- Current measurement
- Power calculation
- Energy calculation
- Data storage
- Threshold comparison
- Alert generation
- Real-time monitoring

C.1 Main Program Flow

Start → Initialize Sensors → Read Voltage → Read Current → Calculate Power → Calculate Energy → Display Data → Compare Threshold → Generate Alert → Repeat

Complete source code can be maintained in the approved project repository, while only essential code excerpts should be included in the report.

Appendix D – Datasheets

Include the relevant datasheets/specifications for:

Component	Information to Include	Specifications / Key Details	Figure (Example)
Voltage Sensor (ZMPT101B)	• Measurement range • Accuracy • Output • Operating voltage	• Measurement range: 80 – 260 VAC • Accuracy: ±0.2% • Output: 0 – 2.5 V • Operating voltage: 5 V DC	 Figure D.1: Voltage Sensor (ZMPT101B)
Current Sensor (ACS712 20A)	• Current range • Sensitivity • Accuracy • Operating voltage	• Current range: -20 A to +20 A • Sensitivity: 100 mV/A • Accuracy: ±1.5% • Operating voltage: 5 V DC	 Figure D.2: Current Sensor (ACS712 20A)
Microcontroller (ESP32)	• Processor • Clock Speed • Flash Memory • ADC • Operating Voltage	• Processor: Dual-core 32-bit • Clock Speed: up to 240 MHz • Flash Memory: 4 MB • ADC: 12-bit, up to 18 channels • Operating Voltage: 3.3 V DC	 Figure D.3: Microcontroller (ESP32)
Display (16x2 LCD)	• Operating Voltage • Interface • Display Type • Size	• Operating Voltage: 5 V DC • Interface: I2C • Display Type: Character LCD • Size: 16x2	 Figure D.4: 16x2 LCD Display
Power Supply (5V Adapter)	• Input • Output	• Input: 230 VAC, 50 Hz • Output: 5 V DC, 2 A	 Figure D.5: Power Supply (5V Adapter)
Other Components	• Resistors, Capacitors, Connectors, Wires, PCB, etc.	• Standard electronic components used for signal conditioning and connections.	 Figure D.6: Other Components

Appendix E – Experimental / Raw Data

This appendix contains the original measurements collected during testing.

Sample No.	Time	Voltage (V)	Current (A)	Power (W)	Energy (Wh)	Alert
1	[Time]	[Value]	[Value]	[Value]	[Value]	Normal
2	[Time]	[Value]	[Value]	[Value]	[Value]	Normal
3	[Time]	[Value]	[Value]	[Value]	[Value]	Normal
4	[Time]	[Value]	[Value]	[Value]	[Value]	High
5	[Time]	[Value]	[Value]	[Value]	[Value]	High

The raw measurements should be retained so that the reported results can be independently checked.

Appendix F – Ethics / Institutional Approval

Where applicable, this appendix can contain:

- Project ethics declaration.
- Data-use statement.

- Safety approval.
- Institutional approval.
- Confirmation that collected data were used only for the intended engineering analysis.
- Statement regarding responsible handling of monitoring data.

Appendix G – Project Meeting / Progress Records

Date	Activity	Discussion / Decision	Progress
[Date]	Topic selection	Finalized project topic	Completed
[Date]	Literature review	Reviewed existing energy monitoring systems	Completed
[Date]	System design	Finalized system architecture	Completed
[Date]	Implementation	Developed monitoring system	Completed
[Date]	Testing	Conducted verification	Completed
[Date]	Report	Prepared final report	Completed

Appendix H – User / Industry Feedback

If feedback was collected, include:

- Feedback from supervisor.
- Feedback from users.
- Suggestions for improving the monitoring interface.
- Suggestions for improving system functionality.
- Observations regarding usability.

If no formal feedback was collected, write:

No formal industry or external user feedback was collected during the current project. Future work may include user evaluation to assess usability and practical effectiveness.

Appendix I – Publications / Patents / Competitions / Product Outcomes

For this project, include relevant outcomes if available:

- Conference paper
- Journal paper
- Poster presentation
- Project competition
- Prototype demonstration
- Product development

- Patent application

If none are applicable:

No publications, patents, competitions, or commercial product outcomes were produced as part of the current project.

Appendix J – Individual Contribution Evidence

Activity	Individual Contribution
Problem identification	Identified the energy monitoring problem
Literature review	Reviewed existing energy monitoring approaches
Requirements analysis	Defined functional and performance requirements
System design	Developed the proposed architecture
Implementation	Worked on data acquisition and processing
Testing	Prepared and conducted system tests
Data analysis	Analysed energy consumption measurements
Documentation	Prepared the project report
Presentation	Prepared project presentation and demonstration

Appendix Summary

The appendices provide supporting evidence for the **Real-time Energy Monitoring and Power Consumption Analysis** project, including calculations, system schematics, source code, component specifications, raw measurements, project records, and individual contribution evidence. This supporting material strengthens the technical validity and traceability of the main report.